$L_D \gg L$, $L_D \gg L_{NL}$, ignore D.

when $L_D \sim L_{NL}$. 需考虑 GVD 与 SPM 联合作用.

—— Soliton.

NLSE: 归一化形式:

$$i\frac{\partial U}{\partial \xi} = sgn(\beta_2)\frac{1}{2}\frac{\partial^2 U}{\partial \tau^2} - N^2 e^{-\alpha z}|U|^2 U.$$

ξ, τ 为归一化距离与时间变量:

$$\xi = z/L_D, \quad \tau = T/T_0$$

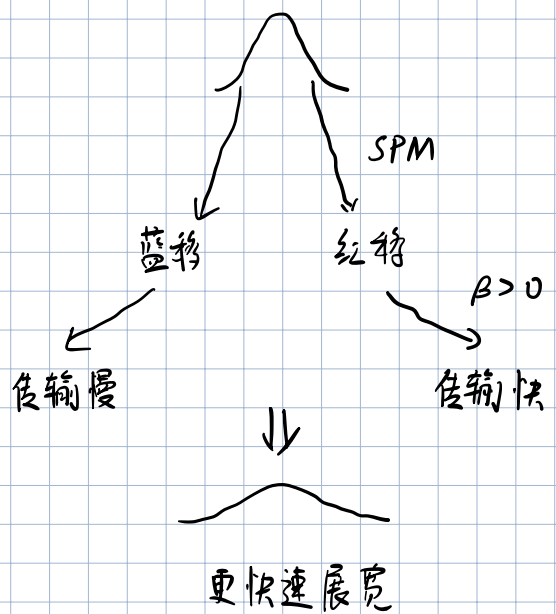
参量 N:

$$N^2 \equiv \frac{L_D}{L_{NL}} = \frac{\gamma P_0 T_0^2}{|\beta_2|}.$$

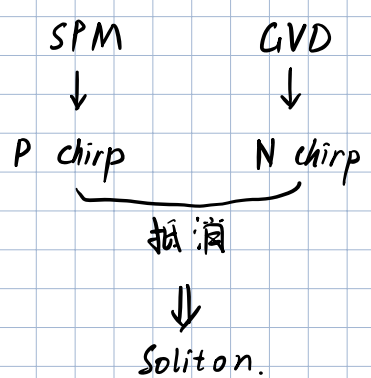
- $N \gg 1$, SPM 主导
- $N \ll 1$, GVD 主导
- $N \sim 1$, 共同主导.

$$sgn(\beta_2) = \begin{cases} +1 & \beta_2 > 0 \\ -1 & \beta_2 < 0 \end{cases}$$

□ $\beta > 0$.



□ $\beta < 0$.



脉冲整形为 双曲正割.

光波分裂.

四波混频?

三阶色散. TOD.

$\lambda_0 \sim \lambda_D$ (零色散波长). $\beta_2 \approx 0$.

引入 TOD 长度 L_D', Def 归一化距离 $\xi' \equiv z/L_D'$

∴

$$i\frac{\partial U}{\partial \xi'} = sgn(\beta_3)\frac{i}{6}\frac{\partial^3 U}{\partial \tau^3} - \bar{N}^2 e^{-\alpha z}|U|^2 U.$$

$$\bar{N}^2 \equiv \frac{L_D'}{L_{NL}} = \frac{\gamma P_0 T_0^3}{|\beta_3|}$$

- $\bar{N} \ll 1$. TOD 主导
- $\bar{N} \gg 1$. SPM 主导.